

Maximizing Guest Retention through Cross-Property Analytics

Advance Data Analytics - Final Project

Group 3:

Gerald Abbey

Ansu Joy

Nihal Vellaramkallingal Sulaiman

Souren Chowdhury



The Business Blind Spot

The Problem

- Revenue loss from late-stage booking cancellations

The Blind Spot

- Guest data is analyzed in silos
- Room bookings are separated from spa, casino, and dining data
- No unified view of guest behavior

Key Question

- Do engaged guests (cross-property spenders) cancel less often?



Unifying the Data in the Cloud

Cloud-based analytics pipeline (Databricks)

Data Sources

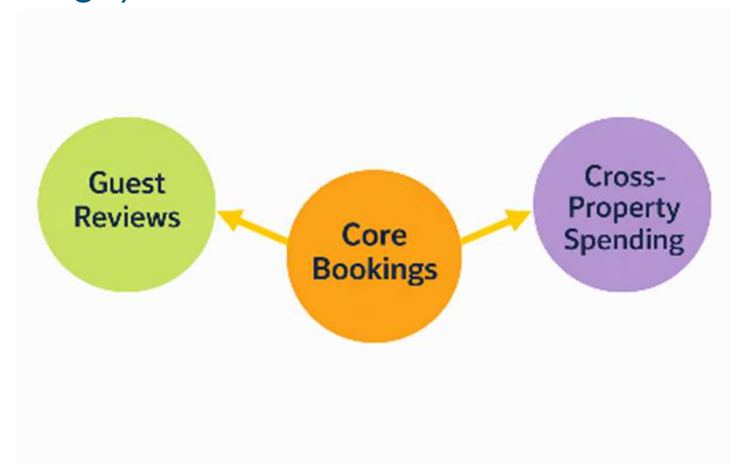
- 119,000+ hotel booking records
- Guest reviews (TripAdvisor dataset)
- Cross-property spending data (synthetic POS logs)

Cloud Integration

- Unified processing on Databricks / GCP
- Scalable data pipeline for large datasets

Outcome

- Unified 360° guest profile in the cloud



Predicting Guest Cancellation Risk

Key Finding

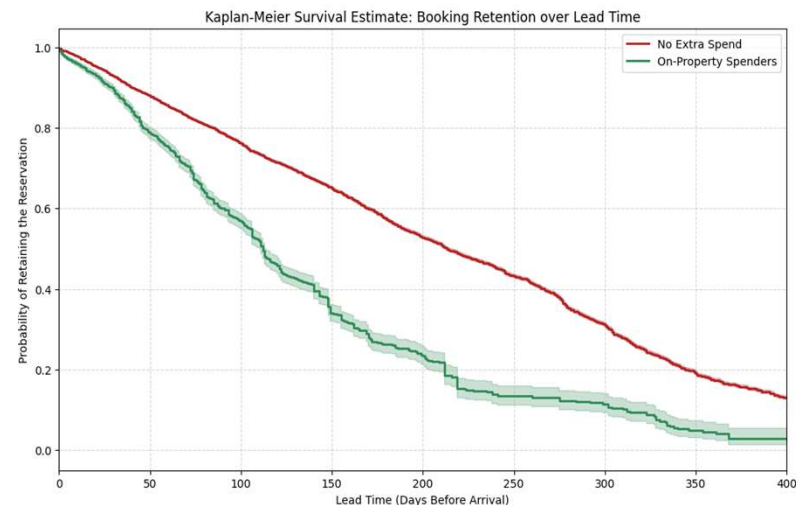
- Guests who engage with on-property amenities are significantly less likely to cancel their reservations.

Survival Analysis Results

- Applied **Kaplan-Meier** for guest retention modelling.
- Compared **On-Property vs Room-Only** guests.
- Identified **high-risk pre-arrival cancellation** windows.
- Cross-property guests showed stronger retention

Business Insight

- Room-only guests are more vulnerable to cancellation.
- The highest cancellation risk occurs approximately **14–30 days before arrival**.
- Early identification enables proactive retention campaigns.
- Personalized offers can be triggered before guests reach peak cancellation risk.



AI-Driven Guest Personalization System

Intelligent Guest Targeting System

- Converts guest feedback into actionable decisions
- Combines sentiment analysis + guest behavior patterns

Automated Actions (An automated NLP recommendation engine)

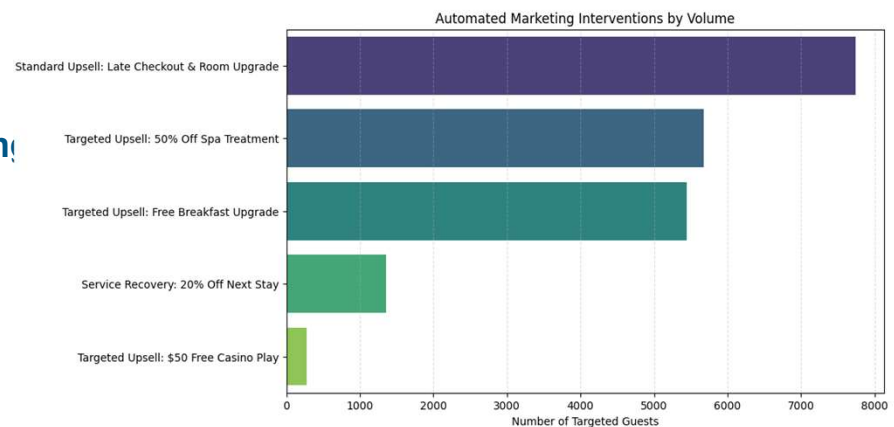
- **Negative experience** → **Service Recovery offer** (discounts, apology incentives)
- **Positive experience** → **Targeted upsell** (spa, casino, dining packages)

How It Works

- Analyzes guest reviews using **sentiment scoring**
- Assigns personalized offers automatically

Goal

Right offer → Right guest → Right time



Business Impact (ROI)

Revenue Protection

- Reduced booking cancellations
- Improved revenue retention

Revenue Growth

- Higher cross-property spending
- Increased conversion from personalized offers

Customer Value

- Improved guest satisfaction
- Higher customer lifetime value (CLV)

Strategy Shift:

- Shift from reactive to predictive revenue management



Recommendations & Next Steps

Recommendations

- Trigger offers before high-risk cancellation periods
- Use guest sentiment for real-time service recovery
- Promote bundled experiences (hotel + spa + casino)
- Integrate system into CRM for automation

Final Goal

- Maximize guest lifetime value across all properties



Thank You

